higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

ELECTRONIC CONTROL AND DIGITAL ELECTRONICS
NQF LEVEL 3

14 NOVEMBER 2019

This marking guideline consists of 6 pages.

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QUESTION 1: FUNDAMENTALS OF ELECTRONICS

1.1 1.1.1
- Testing
- Troubleshooting
- Repairing of electronic equipment (Any 2 × 1) (2)

1.1.2
- Select Ω function on multimeter
- Select appropriate range
- Remove resistor from circuit
- Contact the probes in parallel/across the resistor
- Read value in correct sub-unit (Any 3 × 1) (3)

1.2 1.2.1 Insulator: There is a wide energy gap between the valence and conduction band.
Electrons cannot move easily between bands.

Conductor: There is no energy gap, as the valence and conduction bands overlap.
Electrons move easily between bands. (4)

1.2.2 Covalent bonding (1)

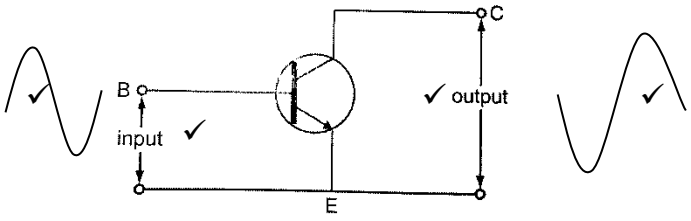
1.3 1.3.1 (a) A – Germanium
(b) B – Silicon
(2 × 1) (2)

1.3.2
- Tunnel diode operates in the reverse bias.✓
- As the reverse bias increases,✓ the capacitance decreases.✓ (3)

1.3.3 (a) V_F – Forward voltage
(b) I_R – Reverse current
(2 × 1) (2)

1.4 1.4.1 $I_C = I_E – I_B$✓
$= 10 – 3$
$= 7$ mA✓ (2)

1.4.2



(4)

1.5 1.5.1 False
1.5.2 False
1.5.3 True
1.5.4 True
1.5.5 False

(5 × 1) (5)

1.6 1.6.1
- At 0 V at the gate of the MOSFET, the channel is open and current can flow through the channel.
- As the gate voltage becomes negative, the channel closes, thus reducing the current flow through the channel.
- If the gate voltage has to become positive, the channel widens, thus allowing more current to flow through the channel.
- With a positive voltage at the gate, the MOSFET acts like an enhancement mode MOSFET. (4)

1.6.2 A spike of voltage Electrostatic Discharge (ESD) will break through the thin layer of metal oxide under the gate area.✓ This will damage the gate control of the component, thus becoming damaged. (2)

1.7 1.7.1 $V_{OUT} = -(R_F \div R_{IN}) \times V_{IN}$✓
$= -(180\ k\Omega \div 30\ k\Omega) \times 0,5$✓
$= -3\ V$✓ (3)

1.7.2 (a) Wave shaping
(b) Impedance matching
(c) Audio amplification

(3 × 1) (3)
[40]

QUESTION 2: DIGITAL ELECTRONICS

2.1 2.1.1 11011_2 to a decimal number

$1 \times 2^4 = 16$
$1 \times 2^3 = 8$ ✓
$0 \times 2^2 = 0$
$1 \times 2^1 = 2$
$1 \times 2^0 = 1$ ✓
+
$= 27$

Answer: $11011_2 = 27_{10}$✓

2.1.2 45_8 to a binary number

4	5
100✓	101✓

100101

Answer: $45_8 = 100101_2$✓

2.1.3 236_{10} to a hexadecimal number

1 6	236		
1 6	14	r 0,75 × 16	= 12 = C ✓
	0	r 0,875 × 16	= 14 = E ✓

Answer: $236_{10} = EC_{16}$✓

(3 × 3) (9)

2.2 2.2.1 Subtract the following binary numbers:
$100010_2 - 11101_2$

				$_2$✓		$_2$✓
	1	0	0	0	1	0
-	0	1	1	1	0	1
	0	0	0	1	0	1

Answer = 101_2 ✓

2.2.2 Multiply the following binary numbers:
$10110_2 \times 1001_2$

				1	0	1	1	0
				×	1	0	0	1
.	.	.	1	0	1	1	0	✓
.	.	0	0	0	0	0	.	
.	0	0	0	0	0	.	.	
1	0	1	1	0	.	.	.	✓
1	1	0	0	0	1	1	0	

Answer = 11000110_2 ✓

2.2.3 Divide the following binary numbers:
$11101_2 \div 110_2$

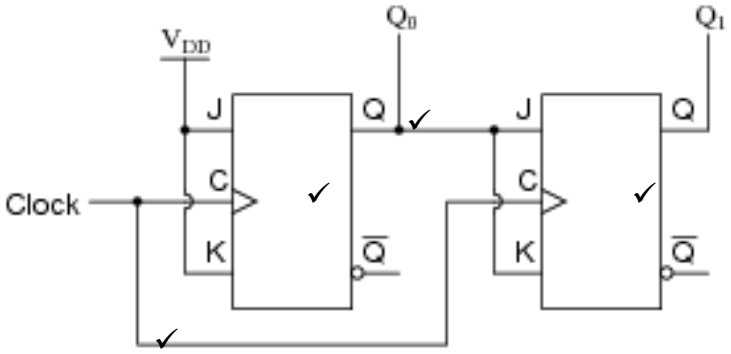
					1	0	0	,1
1	1	0	1	1	1	0	1	0
		-	1	1	0	✓		✓
			0	0	1	0	1	0
					-	1	1	0
						1	0	0

Answer = $100,1_2$ ✓

(3 × 3) (9)

2.3 2.3.1
- The circuit is configured to reset on the 10th trigger.
- The bit counts from 0 to 9.
- It has 10 states of its output. (3)

2.3.2



(4)
[25]

QUESTION 3: ASSEMBLING A PERSONAL COMPUTER

3.1 3.1.1
- Listen for sounds of operation – fan running, beep sounds.
- Look for messages on the monitor.
- Check if computer has power. (3)

3.2 Cache is temporary✓ memory where frequently used instructions and data are stored.✓ (2)

3.3 3.3.1 VGA – monitor or data projector
3.3.2 PCI – installation of expansion cards, for example TV, sound card
(2 × 1) (2)

3.4
- A grounded wrist strap is used so as not to attract static.
- Power off the tower and remove side cover.
- Locate the CPU by identifying its cooling fan.
- Release clips that mount cooling fan and remove fan.
- Release the locking lever on the side of the CPU socket.
- Remove the old CPU IC gently, not to damage the pin set.
- Align the new IC to the socket and press in gently.
- Lock the side lever again to hold IC in place.
- Apply thermal paste on cooling fan and mount into place. (Any 8 × 1) (8)

[15]

QUESTION 4: TRANSDUCERS

4.1 Transducers convert non-electrical✓ energy to electrical energy✓ so as to create electronic control signals. (2)

4.2
- A photosensitive transducer uses a photo emitter and photo receiver to operate.
- Light is reflected or interrupted by the turning disc.
- The received light pulses are converted to electrical pulses and amplified.
- The frequency of the pulses are counted.
- The count is displayed as the speed of the rotating disc. (5)

4.3
- When the moving core of the transformer is in the middle of the two secondary windings, the secondary voltages are equal.
- When the core is moved by an external force or pressure, the secondary voltages differ.
- The value of the secondary voltages gives a reference to the magnitude and direction of force. (3)

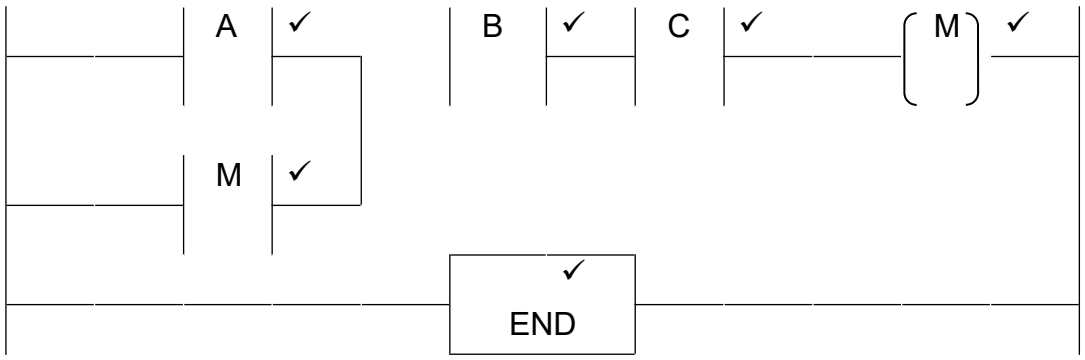
[10]

QUESTION 5: PROGRAMMABLE LOGIC CONTROLLERS

5.1
- Writing programs into the PLC
- Reading existing program on PLC
- Fault-finding
- Simulation (Any 2 × 1) (2)

5.2 5.2.1 Normally closed contact
5.2.2 Normally opened relay/coil
(2 × 1) (2)

5.3



(6)

[10]

TOTAL: 100

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